Experiment – 4: Verification of System Properties: Linearity, Time Invariance, Causality and Stability

Objective :

i) To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses

Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
% Experiment 11: Linearity and Time Invariance
clc;
clear;
close all;

t = -1:0.001:1;
x1 = sin(2*pi*5*t);
x2 = cos(2*pi*5*t);

a = 2;
b = 3;
t0 = 0.2;

T = @(x) 2*x;

% Linearity Test
lhs = T(a*x1 + b*x2);
rhs = a*T(x1) + b*T(x2);
linError = max(abs(lhs - rhs));

% Time Invariance Test
```

```
xDelay = sin(2*pi*5*(t - t0));

lhsTI = T(xDelay);
rhsTI = 2*sin(2*pi*5*(t - t0));

tiError = max(abs(lhsTI - rhsTI));

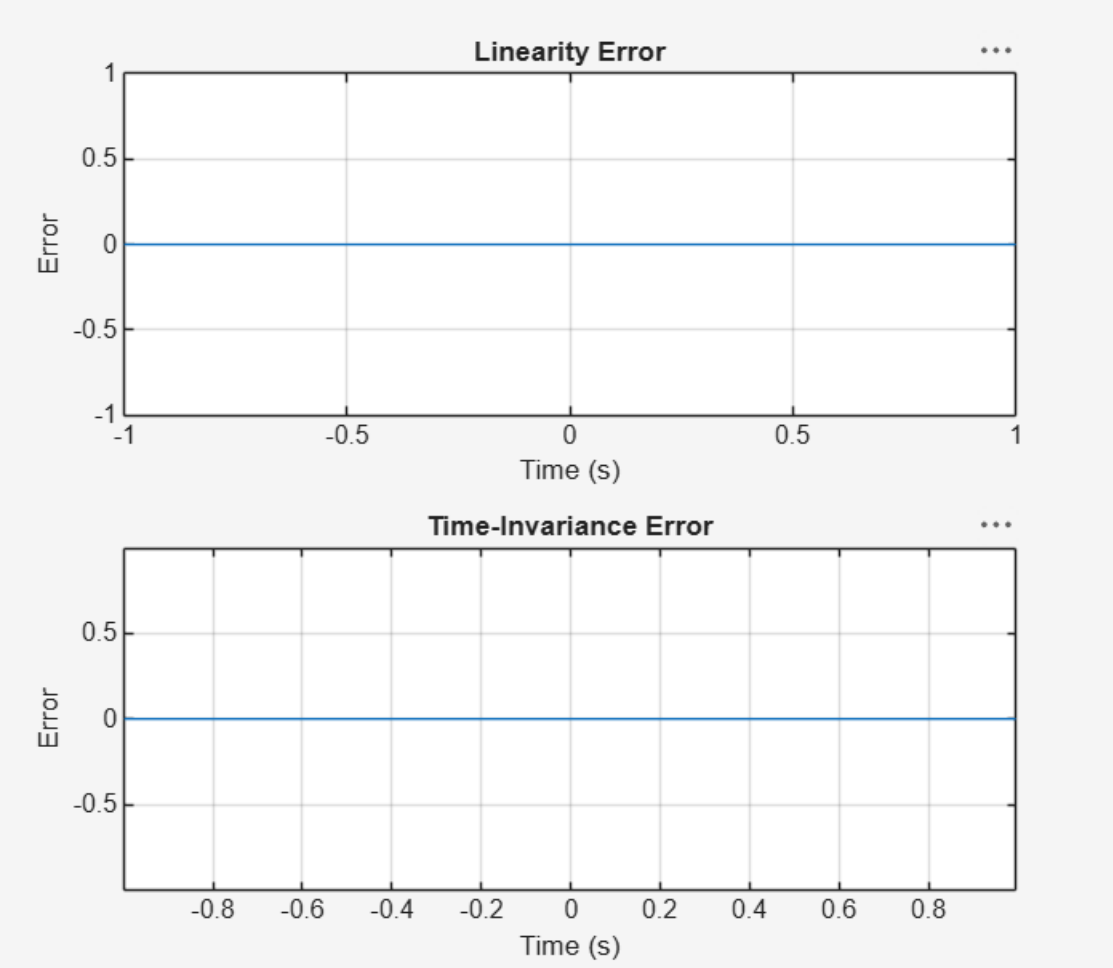
% Plot Results
figure;

subplot(2,1,1);
plot(t, lhs-rhs);
grid on;
title('Linearity Error');
xlabel('Time (s)');
ylabel('Error');

subplot(2,1,2);
plot(t, lhsTI-rhsTI);
grid on;
title('Time-Invariance Error');
xlabel('Time (s)');
ylabel('Error');

fprintf('Linearity error = %g\n', linError);
fprintf('Time-invariance error = %g\n', tiError);
```

OUTPUT:



RESULT:

The system properties were successfully verified and classified. Linearity, time invariance, causality, memory, and stability were analyzed. The results confirmed the system behavior for practical applications.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(i) To verify the linearity and time-invariance properties of continuous-time and discrete-time systems using suitable input signals and system responses.	Linearity Verification (Homogeneity & Additivity)	Correctly applies the homogeneity test and the additivity test (or the combined superposition test) to every CT and DT system. Chooses suitable input signals and scaling constants, compares the two responses accurately, and concludes linear or non-linear	Applies the linearity tests correctly to most systems, with minor errors in the choice of inputs, algebra or comparison.	Performs a basic linearity test (only one of homogeneity or additivity, or only some systems) with noticeable errors, and needs guidance.	Unable to apply or interpret the linearity test.	20	2	

		correctly.						
	Time-Invariance Verification (Time-Shift Test)	**Correctly applies the time-shift test, comparing the response to the shifted input with the shifted output for every system. Handles systems with time-scaling or time-varying coefficients correctly and concludes time-invariant or time-varying correctly.**	**Applies the time-shift test correctly to most systems, with minor errors in shifting or comparison.**	**Performs a basic time-shift test with noticeable errors (for example shifting only the input or only the output), and needs guidance.**	**Unable to apply or interpret the time-invariance test.**	**20**	**2**	

	MATLAB Implementation & Numerical Verification	**Writes correct MATLAB code with suitable time and index vectors to generate the test inputs, the system responses, the scaled, summed and shifted versions, and the plots. Computes numerical errors between the compared responses, and uses a tolerance to decide pass or fail. Plots have titles,**	**Code, plots and numerical comparison are correct, with minor errors in range, parameters, tolerance or labelling.**	**Code runs but has noticeable errors in vectors, shifting or error calculation, and needs assistance.**	**Unable to implement the tests or produce meaningful responses and plots in MATLAB.**	**20**	**2**	

		axis labels and legends.						
	System Classification & Mathematical Justification	**Independently classifies each CT and DT system as linear or non-linear and time-invariant or time-varying Supports each classification with correct mathematical working and with the waveform or error evidence from MATLAB.**	**Classifies most systems correctly, with minor errors or limited justification.**	**Gives basic classifications. Justification is incomplete or not linked to the test results, and needs guidance.**	**Unable to classify the systems correctly or justify the classification.**	**20**	**2**	

	Engineering Analysis, Record & Interpretation	**Provides a technically accurate interpretation of the results and links them to the importance of linear time-invariant (LTI) systems Maintains a complete, well-organized record with code, plots, observations and conclusions.**	**Maintains a proper record with minor omissions. Interpretation is correct but brief.**	**Record and interpretation are basic or partly incomplete, and need guidance.**	**Incomplete record, limited or no interpretation of the results.**	**20**	**2**	
					Total	**100**	**0**	